

YUQI GUO

Software Engineer | Backend, Distributed Systems & AI Infrastructure
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TECHNICAL SKILLS

Languages & Backend Java, Python, JavaScript, SQL | Spring Boot, REST APIs, Microservices, Kafka, RabbitMQ, MCP, RAG
Data, Cloud & Ops PostgreSQL, Snowflake, Redis, OpenSearch, Geode/GemFire | AWS, GCP, Docker, Kubernetes, CI/CD, Prometheus/Grafana/Kibana

PROFESSIONAL EXPERIENCE

Goldman Sachs

Software Engineer

Salt Lake City, UT

Jan 2025 - Present

- Design and delivery of **Spring Boot microservices** for the *Margin Agreement* lifecycle, including creation, validation, amendments, *economic calculations*, document generation, and APIs over deeply nested legal and financial data.
- Built **event-driven workflows and scheduled state-machine orchestration** with Kafka and RabbitMQ; implemented batch consumers, manual acknowledgments, keyed retry/republish, backpressure, and consumer-lag monitoring for reliable processing.
- Developed a **priority-scored rule-tree engine** that evaluates *legal terms*, *regulatory rules*, and *agreement attributes*, returning closest-match diagnostics to reduce manual analysis across *netting*, *enforceability*, *funding*, and *margin decisions*.
- Integrated **Snowflake**, **Elasticsearch/OpenSearch**, and **Apache Geode/GemFire** to support streaming data access, near-real-time search, low-latency reference lookups, and Kibana operational dashboards.

CuraStone Corp

Backend Developer Intern

Remote, United States

Aug 2023 - Dec 2023

- Built a **Spring Boot PDF conversion service** that processed **5,000+ documents in two months**; secured REST APIs with JWT authentication through AWS Cognito.
- Containerized and deployed the service on **AWS ECS** with DynamoDB, improving scalability and deployment consistency.
- Added **JUnit/Mockito tests**, **OpenAPI documentation**, and **automated end-to-end API validation** to improve release reliability and developer productivity.

OPEN-SOURCE & SELECTED PROJECTS

Open-Source Contributions | Kubernetes, Google ADK, LangExtract

Aug 2025 - Present

- Contributed merged **PR #923** to [google/adk-java](https://github.com/google/adk-java), eliminating **linear memory growth and eventual OOM risk** in *InMemorySessionService* by atomically pruning empty parent maps from a three-level *ConcurrentHashMap* while preserving active sessions; added regression tests for complete cleanup and partial retention.
- Contributed merged **PR #385** to [google/langextract](https://github.com/google/langextract), designing and implementing **per-chunk retry isolation** with configurable exponential backoff and retryable-error classification, preventing temporary *Gemini provider throttling and capacity failures* from aborting long-running parallel extraction jobs while preserving progress across unaffected chunks.
- Contributed merged PRs **#4746** and **#4751** to [kubernetes-client/java](https://github.com/kubernetes-client/java), adding a **backward-compatible, regeneration-safe opt-in** for *managedFields deserialization* without changing existing defaults, and restoring the **cert-manager model-generation workflow** through a local container-image fallback when the private registry was unavailable.

AI Infrastructure (Personal Portfolio) | [Demo](#) · [Frontend](#) · [AI Platform](#) · [Admin](#) · [Notification](#) · [Analytics](#) Nov 2023 - Present

- Architected and deployed a **multi-service AI platform** spanning LLM agent orchestration, RAG retrieval, content administration, verified subscription workflows, notifications, search, and real-time behavior analytics.
- Designed a **secure MCP gateway** that routes LLM intents to **schema-validated tools** with RBAC, idempotency, confirmation-gated writes, recoverable conversation state, human handoff, and structured audit logging.
- Engineered **replay-safe event pipelines** using a **Kafka transactional outbox**, consumer groups, manual acknowledgments, retry/DLQ handling, Redis idempotency, and OpenSearch indexing to decouple transactional writes and **prevent duplicate downstream processing**.
- Built a **privacy-aware analytics layer** that reconstructs user sessions and **ordered conversion funnels**, computes engagement and recommendation signals, isolates sensitive raw events, and serves **low-latency aggregate APIs** to web dashboards and AI agents.

EDUCATION

Syracuse University

M.S. in Computer Science, GPA: 3.6/4.0

Syracuse, NY

May 2024

University of Liverpool

B.S. in Computer Science, First Class Honors

Liverpool, UK

Jul 2022